Do Now · retrieval

Lesson 071 · Pipelining and Computational Methods Review

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a heuristic?

Q2

When does backtracking undo a choice?

Q3

What does throughput measure?

Q4

Rising-edge D flip-flop Q=1, D=0 immediately before active edge. What is Q after the edge?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Pipelining overlaps separate stages of different tasks; it increases throughput after filling the pipeline, not necessarily single-task latency.

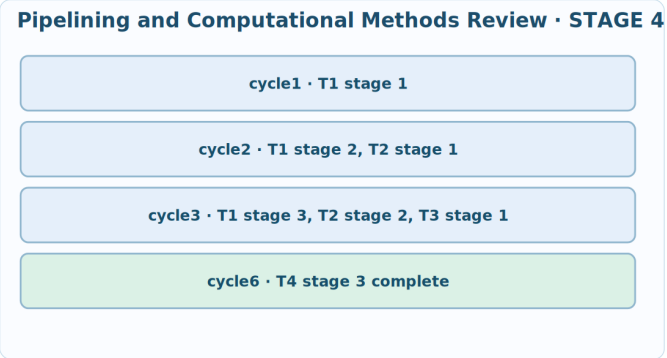
Core idea 2

Explain — Four tasks on a three-stage pipeline complete in six ideal cycles, versus twelve when each task runs all stages in isolation.

Core idea 3

Evaluate — Choose the computational method that fits: decomposition for parts; backtracking for reversible choices; heuristic for guided estimates; visualisation for patterns.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

pipeline — Overlapping multiple processing stages.

throughput — Work completed per time.

latency — Time for one task to finish.

stall — Delay from dependency or resource issue.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

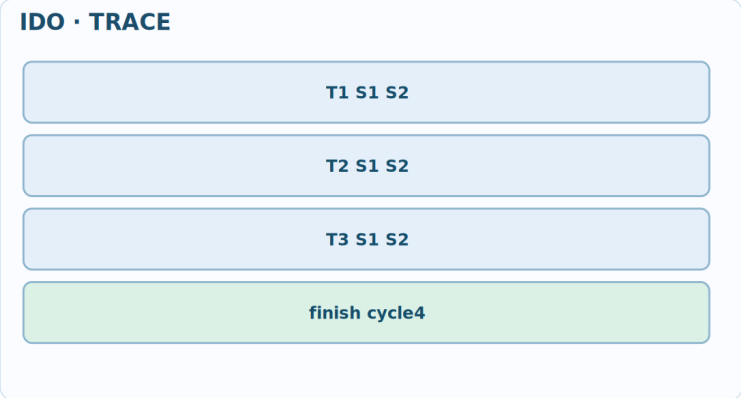
Problem

Find ideal cycles for 3 tasks through 2 single-cycle stages.

Follow the worked method

1. Explain — Start a new task each cycle while earlier tasks advance; last task finishes cycle4.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain
This is a throughput model assuming one task enters each cycle and no stall.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Draw ideal 2-stage pipeline schedule for 3 tasks and state completion cycle. [6]

Question 2

For 6 tasks and 4 one-cycle stages, compare pipelined and sequential ideal times. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Match backtracking, data mining, visualisation and pipelining to four problems: maze with dead ends, shopping patterns, charting results and overlapping video processing. Explain each. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A pipeline always divides each individual task's latency by its number of stages.

Use key vocabulary from memory

1. Explain pipeline in your own words.

2. Explain throughput in your own words.

3. Explain latency in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.